

Virtual Reality – Unit 2 Notes

Input Devices: Trackers, Navigation and Gesture Interfaces

Introduction

- Interactivity is one of the three I's of VR (Immersion, Interaction, Imagination).
- Special interfaces input the user's commands to the computer and give feedback from the simulation to the user.
- Each device addresses a different sense: 3D trackers/sensing suits measure body motion; sensing gloves digitize hand gestures; stereo HMDs and large displays give visual feedback; 3D sound generators give audio.
- Goal of research: faster, more natural interaction that overcomes the communication bottleneck of keyboard and mouse.
- This chapter: tracking, navigation and gesture input. Output devices (visual, auditory, haptic) come in the next chapter.
- VR system architecture: VR Engine $\rightleftharpoons$ Input devices $\rightleftharpoons$ User $\rightleftharpoons$ Task; VR Engine $\rightleftharpoons$ Software & Databases.

2.1 Three-Dimensional Position Trackers

- Used in navigation, robotics, biomechanics, architecture, CAD, education and VR – all need real-time position and orientation of moving objects.
- Requirements vary in range, precision and update rate. GPS is accurate to only 1–5 m (95% of time): fine for car navigation, far too coarse for VR. VR distances are usually no larger than a room or lab.
- A moving object in 3D has 6 degrees of freedom (DOF): 3 translations (X, Y, Z) and 3 rotations – yaw, pitch and roll (about the axes). Six numbers must be measured fast enough.

Definition: A tracker is special-purpose hardware used in VR to measure the real-time change in a 3D object's position and orientation.

- VR tracks the head, hands or limbs for view control, locomotion and object manipulation. Newer use: controlling an avatar (virtual body mapped to the user).

Example: HMD with magnetic tracker (Fig. 2.2)

- Tracker receiver sits on the user's head; a transmitter is fixed in the room.
- Head motion is sampled by an electronic unit and sent to the host (graphics workstation).
- Host computes a new viewing direction, renders an updated image, converts it to NTSC video and shows it on the two LCD screens (left-eye and right-eye video generators).
- Without head tracking the view could not match head posture and the sense of immersion would be lost. The same need applies to larger displays such as the Immersive Workbench.
- 3D sound also uses tracker data: it lets the computer colocate sound sources with virtual objects, increasing realism and immersion. Accuracy needs for sound are much lower than for graphics (visual acuity > auditory localization acuity).
- Competing tracking technologies: mechanical, magnetic, optical, ultrasonic and hybrid. Compare performance parameters to match sensory needs and budget.

2.1.1 Tracker Performance Parameters (Fig. 2.3)

All trackers share four key parameters: accuracy, jitter, drift and latency.

Parameter	Meaning	Notes
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Accuracy	Difference between the object's actual 3D position and that reported by the tracker.	Given separately for translation (fraction of mm) and rotation (fraction of degree). Not constant – degrades with distance from the reference origin. Related terms: resolution, repeatability.
Jitter	Change in tracker output when the tracked object is stationary.	Appears as “noise” – readings scatter around a value instead of staying constant. Not constant over the work envelope; affected by environmental conditions near the tracker.
Drift	Steady increase in tracker error with time.	Output slowly wanders away from the true position of a stationary object. Corrected using a secondary sensor (e.g. inertial drift compensation) – typical of hybrid trackers.
Latency	Time delay between the action and the result; for trackers, the time between the change in object position/orientation and the sensor detecting it.	Minimal latency is desired. Large latency between e.g. head motion and the scene change causes “simulation sickness” (nausea).

- Total (system) latency = tracker measurement delay + communication delay + computer render/display time. The user feels the total, not just tracker latency.
- Measurement, communication and display loops are asynchronous; system latency can be reduced by synchronizing them (generation lock, “genlock”). Genlock reduces but does not eliminate latency.
- Fast communication lines and streaming mode (sensor data sent continuously) reduce latency – best for fast-moving objects.

2.2 Navigation and Manipulation Interfaces

Definition: A navigation/manipulation interface is a device that allows interactive change of the view of the virtual environment and exploration through selection and manipulation of a virtual object of interest.

Absolute vs Relative Control

	Absolute	Relative
Reference	Position/orientation relative to a fixed coordinate system (transmitter-fixed).	Incremental change relative to the object's previous 3D position.
Output at rest	Never a zero set, even at rest.	Always returns zeros if not acted upon.
Mapping	VR object position mapped directly to the receiver's absolute position.	Object incremented by six signed quantities every simulation cycle; velocity controlled indirectly (larger increment = higher velocity).
Examples	Trackers, 3Ball, 3D Mouse	Trackballs (Magellan), Cubic-Mouse indexed motion

2.2.1 Tracker-Based Navigation/Manipulation Interfaces

- Trackers housed in a structure with user-programmable pushbuttons become navigation/manipulation interfaces.
- **3Ball** – Polhemus 3Ball: hollow billiard ball with tracker receiver inside and a pushbutton on the surface. Use: moves a virtual camera along the 3D vector controlled by the ball while the button is pressed.
- **Wand** – Wand: projects a virtual ray along the tracker-controlled vector. Objects hit by the ray are selected with the button, then repositioned by wrist motion.
- **3D Mouse** – Ascension Technology 3D Mouse: works as a normal 2D mouse and as a 3D navigation/manipulation device; extra functions through its three programmable buttons.

Problem of Absolute Position Control

- Limited arm reach – worse with larger displays.
- Fix 1: multiply tracker readings by a gain > 1 . Drawback: tracker noise is amplified, so objects look jittery.
- Fix 2: indexed motion – object is selected/deselected repeatedly. When deselected it stays in place while the user moves the arm back, then reselects and moves it again (a “clutch”).

The Cubic-Mouse (Fig. 2.21)

- Plastic cube $9 \times 9 \times 9$ cm containing a Polhemus Fastrack tracker, three mutually perpendicular translating rods, and six application-programmable buttons (on the “up” face; wires exit through the “down” face).
- Designed for manipulating a single large virtual model (not fly-by navigation like wands): rotate/translate the model with the tracker, dissect/edit it with the rods.
- Each rod controls a cutting plane; buttons at rod ends select cutting-plane orientation. Control buttons: one clutch button (indexed motion) and two zoom in/out buttons.
- Dual-hand operation: cube held in the non-dominant hand, rods/buttons operated by the dominant hand. Proprioception and tactile cues allow use without looking at the rods.

2.2.2 Trackballs

- Interfaces for navigation/manipulation in relative coordinates, e.g. Logitech Magellan (Fig. 2.22).
- A sensorized cylinder measuring three forces and three torques applied by the hand on a compliant element (spring deformation law).
- Working: fixed central part with six LEDs; a moving outer cylinder with six photo sensors. Forces/torques change photo-sensor output; data sent to host over an RS232 serial line.
- Software gains multiply the values to give differential change in object position/orientation. Larger gain = higher speed, but motion is not smooth if the host cannot refresh fast enough.
- Alternative: force control – measured forces control forces the VR object applies on the simulated environment. Can also be used for fly-by (sensor sets velocity/orientation of a virtual camera).
- Pushbuttons (binary on/off, programmable): increase gain/speed, couple/decouple trackball and VR object, start/stop or reset simulation to default start location.
- Drawback – sensor coupling: pure finger forces cause nonzero torques, so the object may rotate when only translation is wanted.
- Fixes: software filters that read only forces, or hardware filter buttons selecting translation-only, rotation-only or dominant-motion input.

2.2.3 Three-Dimensional Probes

- Need: an intuitive, inexpensive I/O device allowing absolute or relative control. Example: Immersion Probe (early 1990s), later renamed MicroScribe 3D, also used to digitize objects.
- Small sensorized mechanical arm on a support base (6 in × 6 in footprint) with six rotary joints (0–5) = 6 DOF, allowing simultaneous positioning and orientation of the tip.
- Counterbalance near the base reduces user fatigue. Tip position is computed by direct kinematics from joint sensor values and link lengths; host reads joints over RS232.
- Foot pedal (binary switch): select/deselect virtual objects, start/stop navigation, or mark points on a real surface for digitizing. It can also control flying speed; velocity direction = probe tip (last link) orientation.
- Workspace: 50-in.-radius hemisphere (models 3D, 3DX) to 66-in.-radius hemisphere (3DL, 3DLX).
- Mechanical, so unaffected by magnetic fields; over 1000 positions/sec with minimal latency.
- Accuracy errors come from joint sensor resolution, temperature variation and joint tolerances. Open kinematic chains accumulate error from base to tip, so larger models are less accurate. Needs calibration (tip placed in a housing near the base and readings compared with stored joint angles).
- Model 3DX: tip accuracy 0.23 mm (0.009 in.); larger 3DLX: 0.30 mm (0.012 in.).

2.3 Gesture Interfaces

- Trackballs and 3D probes are simple, compact and quiet, but restrict hand motion to a small area near the desk, so natural hand motion is sacrificed and interaction is less intuitive.
- For large gesture-based interaction, devices must keep hand freedom of motion within a volume and ideally sense individual finger motions.
- Finger DOF: flexion–extension and lateral abduction–adduction (palmar/radial abduction). The thumb also has anteposition–retroposition (opposition to the palm) (Fig. 2.24).

Definition: Gesture interfaces are devices that measure the real-time position of the user's fingers (and sometimes wrist) to allow natural, gesture-recognition-based interaction with the virtual environment.

Sensing Gloves

- Most gesture interfaces are sensing gloves with embedded sensors measuring each finger's position versus the palm.
- Gloves differ in: sensor type, number of sensors per finger (one or several), sensor resolution, sampling rate, and tethered vs wireless.
- Commercial examples: Fakespace Pinch Glove, 5DT Data Glove (Fifth Dimension Technology), Didjiglove, Immersion CyberGlove. They measure some or all finger joint angles; some have built-in trackers to also measure wrist motion.

Glove vs Trackball Work Envelope (Fig. 2.25)

- Sensing glove work envelope is much larger than that of trackballs or joysticks (hand moves freely in front of the user, with transmitter nearby, versus a desk-top trackball).
- Trackballs and 3D probes give single-point interaction; gloves give dextrous, multipoint interaction at fingertips or palm – more realistic, especially for object manipulation.
- Gloves can also be navigation interfaces using user-programmed gesture libraries.

Quick Revision Summary

Device	Type	Key point
Tracker (HMD)	Absolute	6 DOF; needs low latency, accuracy; drift/jitter issues

3Ball / Wand / 3D Mouse	Absolute	Tracker + buttons; camera movement, ray selection
Cubic-Mouse	Absolute + indexed	9 cm cube, 3 rods, dual-hand, model dissection
Magellan trackball	Relative	3 forces + 3 torques, RS232, sensor coupling
MicroScribe 3D probe	Absolute/relative	6-joint arm, foot pedal, needs calibration, 0.23 mm accuracy
Sensing gloves	Gesture	Finger joint angles, large work envelope, multipoint